Shift-Invariance and Adversarial Robustness

A Collaborator Primer: background, problem-style proofs, and how to find the next question

companion to the paper and the technical report; for a new theory collaborator

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A guided path that takes a strong undergraduate or new graduate student from zero to (i) reading the paper and the technical report end to end, (ii) reconstructing every proof as a sequence of problems they solve themselves, and (iii) learning the harder skill: how to find the questions, not only follow the steps. Plan three to four weeks of focused study. Companion to `paper/report/main.tex` *(the paper) and* `paper/report/theory_full_checked_v2.tex` *(the technical report, every proof in full).*

Preface

Who this is for. A new collaborator joining the shift-invariance/adversarial-robustness project, mostly on the theory side and a little on the experiments. We assume one undergraduate course each in linear algebra, real analysis or advanced calculus, and probability, plus some exposure to machine learning. You do *not* need prior background in adversarial robustness, Fourier analysis on groups, or implicit-bias theory; all three are built from first principles in Part I.

What this is, and what makes it different. Most primers decompose each proof into steps and ask you to follow them. Following steps is the easy half of mathematics. The hard half, the half that actually produces research, is *coming up with the steps yourself*: knowing which question to ask, which object to define, which special case to test first. This document trains both. Every result appears first as a **Problem** you attempt cold, then a graded ladder of **Hints** you uncover only when stuck, then a full **Solution**. Interleaved are two recurring devices:

- **Research move** boxes that name the general technique a proof secretly used ("quotient out the nuisance," "diagonalize the symmetry," "find the degenerate case"), so you can reuse it.
- **Find-the-question** prompts that hand you a situation with *no question attached* and ask you to generate the right one before reading on. This is the skill that does not come from following steps.

How to use it. Read Part I in order; do the problems before reading their solutions, and give each at least fifteen honest minutes before the first hint. Part II is the paper's theory, result by result; by the end you should be able to state and prove each result with the book closed. Part III is the discovery layer; do it *after* Part II, because it reflects on how the Part II results were found. Part IV points at the open problems where you can contribute and how to run the experiments.

House writing rules (we follow them, so should you). Plain language. Define every term on first use. No em-dashes, no "win"/"beat" as verbs, no AI-tell filler ("delve," "leverage," "crucial"), one spelling convention per document, captions that match figures, honest scoped claims. A claim you cannot verify against the paper, the technical report, or a results file is a claim to verify, not to repeat.

One running thread. A single object recurs from Part I to Part III: the cyclic group of circular shifts acting on a length-d signal, its *orbit* (the set of all shifts of one signal), and the split of $\mathbb{R}^d$ into the constant "DC" line and its zero-sum complement. If you understand that one picture deeply, everything else is commentary on it.

Contents

1 Master notation table

The symbols below are used throughout. Section references point to where each is introduced.

Symbol	Meaning
$x \in \mathbb{R}^d$	input signal or flattened image; d coordinates (§2)
$y \in \{\pm 1\}$	binary label; multiclass uses the active margin M (§2)
$f : \mathbb{R}^d \to \mathbb{R}$	score; prediction $\operatorname{sign} f(x)$ (§2)
$M(x) = F_y(x) - \max_{c \neq y} F_c(x)$	active logit margin (positive iff correct) (§2)
$B_p(\varepsilon)$, $r_p(f, x)$	threat ball of radius ε; robust radius (distance to boundary) (§2)
L, α, $\kappa = L/\alpha$	upper-Lipschitz, co-Lipschitz constants; their ratio (§2, §7)
η	feature-space margin of an invariant feature (§7)
η/L	margin-to-Lipschitz ratio; the certificate scalar (§7)
$G = \mathbb{Z}_d$, S_s	cyclic group of circular shifts; the shift-by-s operator (§3)
$\mathcal{O}(x) = \{S_s x\}$	orbit of x (all its shifts) (§3)
$\hat{x}_k$, $\lvert\hat{x}_k\rvert^2$	unitary DFT coefficient; power spectrum (§3)
$B_{k_1,k_2}(x)$	bispectrum, a degree-three shift invariant (§3)
$V_{\text{inv}} = \operatorname{span}(\mathbf{1})$, V_{ac}	DC (constant) line and its zero-sum complement (§3)
$\Pi_{\text{inv}} = ee^\top$, $e = d^{-1/2}\mathbf{1}$	projector onto the DC line (§3)
$P_G = \lvert G\rvert^{-1} \sum_g R_g$	group-average (Reynolds) projector onto the fixed subspace (§3)
$f_{\text{dc}}(x) = d^{-1/2} \sum_i x_i$	unit DC functional (§3)
γ_{inv}, Sep	invariant linear margin; separation of two sets (§4, §6)
$Q(x) = (\lvert\hat{x}_k\rvert^2)_k$	power-spectrum feature map (§8)
$f_\theta(x) = \sum_r a_r \sigma(w_r^\top x)$	two-layer ReLU network, degree-2 homogeneous (§5)
$\mathcal{H}_K$, Π_G	RKHS of the tangent kernel; its orbit-averaging projection (§12)

Part I: Background you need before the paper

This part assembles the three toolkits the paper rests on: adversarial robustness, Fourier analysis on the cyclic group, and margin/implicit-bias theory. Each subsection states the primitives, works a numerical example you can check by hand, and ends with problems. Skim a subsection if you already know it, but do its problems; they are chosen to expose the exact facts Part II reuses.

2 Adversarial robustness from scratch

The object of study. A classifier is a score $f : \mathbb{R}^d \to \mathbb{R}$ predicting $\operatorname{sign} f(x)$ (multiclass: a vector F predicting $\arg\max_c F_c$). Standard training maximizes clean accuracy. The robustness question asks something stricter: if an adversary may move x by a small vector δ, can it flip the prediction?

Definition 2.1 (Threat model, robust radius, robust accuracy)**.** An ℓ_p threat of budget ε allows $\delta \in B_p(\varepsilon) = \{\delta : \|\delta\|_p \leq \varepsilon\}$. The two standard norms are ℓ_2 (Euclidean) and ℓ_∞ (each pixel moves by at most ε). The *robust radius* at a correctly classified (x, y) is $r_p(f, x) = \inf\{\|\delta\|_p : \operatorname{sign} f(x + \delta) \neq y\}$, the distance to the decision boundary. The *robust accuracy at* ε is the fraction of test points with $r_p > \varepsilon$.

The radius is the finer quantity; robust accuracy is the radius thresholded at ε, and it shows floor effects (everything reads zero) when ε is large relative to typical radii. The paper uses the radius as the dataset-comparable measure and robust accuracy only where a standard threshold makes it meaningful (after adversarial training).

Attacks versus certificates. An *attack* (PGD, AutoAttack, the minimum-norm DDN attack) searches for a small δ and so *upper bounds* the radius: it can only show the model is at least this weak. A *certificate lower bounds* the radius: it guarantees the model is at least this strong. The two bracket the truth from opposite sides. The single most important certificate is the next definition.

Definition 2.2 (Lipschitz–margin certificate)**.** A score g is *L-Lipschitz* toward the boundary if $|g(x)-g(z)| \le L\|x-z\|$ along the relevant paths. If g is correct at x with score margin $\mu = y\,g(x) > 0$, then no δ with $\|\delta\| < \mu/L$ can drive g to zero, so $r(g,x) \ge \mu/L$. For a differentiable score the first-order L is the dual-norm gradient $\|\nabla_x M(x)\|_q$ (q dual to the threat p: ℓ_∞ pairs with ℓ_1, ℓ_2 with ℓ_2).

Worked example 2.3 (A one-line certificate)**.** Let $g(x) = w^\top x$ be linear, correct at x with $\mu = y\,w^\top x$. Then g is exactly $\|w\|_2$-Lipschitz in ℓ_2, so $r_2 \ge \mu/\|w\|_2$. In fact for a linear score this is an equality: the distance from a point to a hyperplane is exactly $\mu/\|w\|_2$. Hold onto that "equality for the linear case"; §7 shows it is the boundary case of a more general two-sided bound.

Problem 2.1 (Why the dual norm)**.** Show that for a differentiable score the largest first-order change $|g(x+\delta) - g(x)|$ over $\|\delta\|_p \le \varepsilon$ is $\varepsilon\,\|\nabla g(x)\|_q$ with $1/p + 1/q = 1$, and deduce the first-order radius $M(x)/\|\nabla M(x)\|_q$.

Hints (peek one at a time):

- First order, $g(x+\delta) - g(x) \approx \nabla g(x)^\top \delta$.
- Maximizing a linear functional over an ℓ_p ball is the definition of the dual (ℓ_q) norm.
- For ℓ_∞ the maximizer puts $\delta_i = \varepsilon\,\mathrm{sign}(\partial_i g)$, giving $\varepsilon\|\nabla g\|_1$.

Solution. By Cauchy–Schwarz's ℓ_p/ℓ_q generalization (Hölder), $\sup_{\|\delta\|_p \le \varepsilon} \nabla g^\top \delta = \varepsilon\|\nabla g\|_q$, attained at the aligning δ. The prediction flips when the change cancels the margin, i.e. at $\varepsilon = M(x)/\|\nabla M(x)\|_q$ to first order. The pairing $\ell_\infty \leftrightarrow \ell_1$ is why an ℓ_∞ attack is read with the ℓ_1 gradient norm, a fact the adversarial-training experiments lean on heavily. □

Research move: Match the dual to the threat

A robustness statement is meaningless until the norm is fixed, and the right companion of an ℓ_p attack is the ℓ_q gradient. Whenever you see $\|\nabla M\|$ in this project, ask "in which norm, and dual to which threat?" The paper's adversarial-training result lives or dies on this matching: the ℓ_1 gradient predicts ℓ_∞ robustness, the ℓ_2 gradient predicts ℓ_2 robustness.

3 Group actions, Fourier analysis, and invariance

Shifts as a group. The cyclic group $G = \mathbb{Z}_d$ acts on $\mathbb{R}^d$ by circular shifts $(S_s x)_i = x_{(i-s) \bmod d}$. Each S_s is an orthogonal permutation (it just renumbers coordinates). The *orbit* of x is $\mathcal{O}(x) = \{S_s x : s \in \mathbb{Z}_d\}$, the set of all d shifts of x. A set is *orbit-closed* if it contains the full orbit of each of its members; a function is *G-invariant* if $f(S_s x) = f(x)$ for all s.

Why Fourier. All shift operators S_s commute and are simultaneously diagonalized by the discrete Fourier transform (DFT). With the unitary convention $\hat{x}_k = d^{-1/2} \sum_j x_j e^{-2\pi ijk/d}$, a shift multiplies coefficient k by the unit-modulus phase $e^{-2\pi isk/d}$. Two consequences run through the whole paper:

- The *power spectrum* $|\hat{x}_k|^2$ is shift-invariant (the phase cancels in modulus) but *phase-blind*.
- The *bispectrum* $B_{k_1,k_2}(x) = \hat{x}_{k_1}\hat{x}_{k_2}\overline{\hat{x}_{k_1+k_2}}$ is also shift-invariant (the three shift phases cancel because $k_1 + k_2 - (k_1 + k_2) = 0$) and, unlike the power spectrum, keeps phase, so it separates signals the power spectrum confuses.

Definition 3.1 (Fixed subspace and the group-average projector)**.** The fixed subspace is $V^G = \{v : R_g v = v \ \forall g\}$ and the *group-average (Reynolds) projector* is $P_G = |G|^{-1} \sum_g R_g$, the orthogonal projector onto V^G. For cyclic shifts the only vectors fixed by every shift are the constants, so $V^G = V_{\text{inv}} = \text{span}(\mathbf{1})$ is the "DC line," $P_G = \Pi_{\text{inv}} = ee^\top$ with $e = d^{-1/2}\mathbf{1}$, and $V_{\text{ac}} = V_{\text{inv}}^\perp = \{v : \sum_i v_i = 0\}$ is the zero-sum "AC" subspace.

Worked example 3.2 (Hand-check at $d = 4$)**.** Take $x = (1, 0, 0, 0)$. Its orbit is the four standard basis vectors. $f_{\text{dc}}(x) = \frac{1}{2}\sum_i x_i = \frac{1}{2}$ for every orbit member, so the DC coordinate is constant on the orbit (it must be: it is shift-invariant). The power spectrum of x is $|\hat{x}_k|^2 = 1/4$ for all k (a spike is flat in frequency), so the power spectrum cannot tell x from any other unit spike, and in particular not from $-x$ shifted. This single fact is why the localized "dot" example of the base paper is the hardest case for invariance.

Problem 3.1 (The projector is the average and it cannot lengthen a vector)**.** Prove that $P_G = |G|^{-1} \sum_g R_g$ is (i) idempotent, (ii) self-adjoint, hence an orthogonal projector, and (iii) non-expansive: $\|P_G v\| \le \|v\|$. Then specialize to cyclic shifts to show $P_G = ee^\top$.

Hints (peek one at a time):

- For (i) use the group property: $R_g P_G = P_G$ because $g \mapsto gh$ permutes the group.
- For (ii) use that each R_g is orthogonal, so $R_g^\top = R_{g^{-1}}$, and summing over the group is symmetric.
- An orthogonal projector never increases the norm by Pythagoras: $\|v\|^2 = \|P_G v\|^2 + \|(I - P_G)v\|^2$.
- For cyclic shifts, $\frac{1}{d}\sum_s S_s$ sends every coordinate to the average, i.e. the matrix $\frac{1}{d}\mathbf{1}\mathbf{1}^\top = ee^\top$.

Solution. (i) $P_G^2 = |G|^{-2} \sum_{g,h} R_g R_h = |G|^{-2} \sum_{g,h} R_{gh} = |G|^{-1} \sum_k R_k = P_G$, since for fixed g, $h \mapsto gh$ is a bijection of G. (ii) $R_g^\top = R_g^{-1} = R_{g^{-1}}$, so $P_G^\top = |G|^{-1} \sum_g R_{g^{-1}} = P_G$. A self-adjoint idempotent is an orthogonal projector; its range is exactly V^G because $P_G v = v \Leftrightarrow R_g v = v \ \forall g$. (iii) Pythagoras gives $\|v\|^2 = \|P_G v\|^2 + \|(I - P_G)v\|^2 \ge \|P_G v\|^2$. For cyclic shifts $\frac{1}{d}\sum_s S_s$ has every entry $1/d$, i.e. $ee^\top$, projecting onto $\text{span}(\mathbf{1})$. □

Research move: Diagonalize the symmetry

Whenever one symmetry group acts, look for the basis that diagonalizes it all at once. For commuting shifts that basis is the Fourier basis, and "invariant" becomes "depends only on magnitudes," "equivariant" becomes "multiply each frequency by a phase." Half the theorems in this project are one line once you are in Fourier coordinates and three pages otherwise. The non-expansiveness in Problem 3.1(iii) returns, unchanged, as the engine of the lazy-regime theorem in §12.

Find-the-question: a phase-blind invariant You are told the power spectrum is shift-invariant but cannot separate a spike from its reflection. Before reading §8, ask: *is there a shift-invariant feature that does keep phase?* What is the lowest-degree polynomial in the $\hat{x}_k$ whose phase factors cancel under a shift? Write the condition $k_1 + k_2 - k_3 = 0$ and see what degree-three monomial it forces. You will have re-derived the bispectrum.

4 Margins, convex hulls, and the max-margin classifier

Geometric margin. For a linear score $w^\top x + b$, the geometric margin at x is $y(w^\top x + b)/\|w\|_2$, the signed Euclidean distance from x to the hyperplane. The hard-margin classifier of two finite sets maximizes the smallest margin.

Theorem 4.1 (Convex-hull duality, classical)**.** *The maximum margin separating two finite sets X_+, X_- equals $\frac{1}{2}\,\mathrm{dist}(\mathrm{conv}\, X_+, \mathrm{conv}\, X_-)$, half the distance between their convex hulls, attained by the hyperplane that perpendicularly bisects the closest pair of hull points.*

We will not reprove this classical fact, but we will use it constantly: it turns "what is the best linear separator" into "how far apart are the two clouds, after whatever transformation we applied." Every margin computation in Part II is an application of Theorem 4.1 to a transformed dataset (projected onto the DC line, or pushed through the power-spectrum map).

Problem 4.1 (One-dimensional margins)**.** Two sets of real numbers, $X_+ \subset [a_+, b_+]$ and $X_- \subset [a_-, b_-]$ with $b_- < a_+$. Show the max-margin threshold sits at $(a_+ + b_-)/2$ with margin $\frac{1}{2}(a_+ - b_-)$, i.e. half the gap between the nearer endpoints.

Hints (peek one at a time):

- On the line, convex hulls are intervals; their distance is the gap between nearer endpoints.
- Apply Theorem 4.1 with $\mathrm{dist}([a_-, b_-], [a_+, b_+]) = a_+ - b_-$.

Solution. The hulls are $[a_-, b_-]$ and $[a_+, b_+]$; since $b_- < a_+$ they are disjoint with distance $a_+ - b_-$, realized by the endpoints b_- and a_+. Theorem 4.1 gives margin $\frac{1}{2}(a_+ - b_-)$ and threshold at the midpoint $(a_+ + b_-)/2$. This is the exact computation behind the invariant linear margin in §6, where the line is the DC axis. □

5 Implicit bias: homogeneous nets, NTK, rich versus lazy

The selection problem. On separable data, infinitely many parameter settings fit perfectly. Which one does gradient descent pick? For a linear model on separable data it converges in direction to the max-margin separator (Soudry et al.). For a positively homogeneous network ($f_{c\theta} = c^k f_\theta$; a bias-free two-layer ReLU net is degree-2), normalized gradient flow converges in direction to a Karush–Kuhn–Tucker (KKT) point of the margin program $\min \frac{1}{2}\|\theta\|^2$ s.t. $y_i f_\theta(x_i) \geq 1$ (Lyu–Li, Ji–Telgarsky). These results say "some KKT point"; *which* one when several exist is the open content the project's optimization section is about.

Two training regimes. A very wide network in the *lazy* (neural-tangent-kernel, NTK) regime stays close to its linearization at initialization and behaves like a fixed kernel machine (Jacot et al.; Chizat et al.). In the *rich* (small-initialization) regime it learns features. The lazy regime is where we can prove things cleanly (§12); the rich regime is where the hardest open question lives.

Feature averaging. On data with many nearly orthogonal cluster centers, the gradient-descent max-margin solution aligns with a signed average of those centers (Frei et al.; Li et al.). That classifies cleanly but is fragile along the individual feature directions, giving a small robust radius. Keep this picture: it is the mechanism the project must either invoke or rule out on orbit-structured data.

Problem 5.1 (Homogeneity and the min-norm representation of one ReLU unit)**.** A single ReLU unit computes $a\,\sigma(w^\top x)$ with $\sigma(t) = \max(t, 0)$. Show the function is unchanged under $(a, w) \mapsto (a/c,\, cw)$ for $c > 0$, and that the representation cost $\frac{1}{2}(a^2 + \|w\|^2)$ is minimized at $c^2 = |a|/\|w\|$, with minimum value $|a|\,\|w\|$.

Hints (peek one at a time):

- σ is positively homogeneous of degree one: $\sigma(c\,t) = c\,\sigma(t)$ for $c > 0$.
- So $a\,\sigma(w^\top x) = (a/c)\,\sigma((cw)^\top x)$. The function is invariant; only the cost changes.
- Minimize $\frac{1}{2}(a^2/c^2 + c^2\|w\|^2)$ over c; set the derivative to zero.

Solution. By degree-one homogeneity $(a/c)\sigma((cw)^\top x) = (a/c)\,c\,\sigma(w^\top x) = a\sigma(w^\top x)$, so the function is unchanged. The cost $\frac{1}{2}(a^2/c^2 + c^2\|w\|^2)$ is convex in c^2 with minimum where $a^2/c^4 = \|w\|^2$, i.e. $c^2 = |a|/\|w\|$, giving value $\frac{1}{2} \cdot 2|a|\|w\| = |a|\,\|w\|$. This "$|a|\,\|w\|$ per unit" bookkeeping is the entire engine of the self-symmetrization theorem in §10; the proof there is this problem applied to a whole orbit of tied units. □

Part II: The paper's theory, result by result

Each section states a result, hands it to you as a problem with a hint ladder, then gives the full solution. Do the problem first. The technical report `theory_full_checked_v2.tex` has the same proofs in reference form; use it to check, not to substitute for the attempt.

6 The invariant linear margin equals a projection separation

The first structural result answers "how much separation survives if the classifier must be shift-invariant?"

Theorem 6.1 (Invariant linear margin)**.** *A linear score $g(x) = w^\top x + b$ is shift-invariant for all x iff $w \in V_{\mathrm{inv}} = \mathrm{span}(\mathbf{1})$. Consequently, for any finite separable classes X_+, X_-, the invariant linear max-margin equals $\gamma_{\mathrm{inv}} = \frac{1}{2}\,\mathrm{dist}(I_+, I_-) = \frac{1}{2}\,\mathrm{dist}(\mathrm{conv}\,\Pi_{\mathrm{inv}}X_+, \mathrm{conv}\,\Pi_{\mathrm{inv}}X_-)$, where $I_\pm = [\min_{x\in X_\pm} f_{\mathrm{dc}}(x),\, \max_{x\in X_\pm} f_{\mathrm{dc}}(x)]$ are the intervals of DC values.*

Problem 6.1 (Prove Theorem 6.1)**.** Two parts. (a) Characterize which w give a shift-invariant linear score. (b) Reduce the invariant max-margin to a one-dimensional problem and evaluate it.

Hints (peek one at a time):

- (a) Invariance for all x means $w^\top S_s x = w^\top x$ for all x, s; turn "for all x" into a condition on w.
- (a) $w^\top S_s x = (S_s^\top w)^\top x$, so you need $S_s^\top w = w$ for all s. Which vectors are fixed by every shift?

- (b) With $w = c\mathbf{1}$, the score depends on x only through $f_{\mathrm{dc}}(x)$. The data collapse to scalars.

- (b) Now it is Problem 4.1 on the DC axis.

Solution. (a) Invariance $w^\top S_s x = w^\top x$ for all x is equivalent to $(S_s^\top w - w)^\top x = 0$ for all x, hence $S_s^\top w = w$ for every s; the only vectors fixed by all cyclic shifts are constants, so $w \in \mathrm{span}(\mathbf{1})$. (b) Writing $w = c\mathbf{1}$, the score is a function of $f_{\mathrm{dc}}(x) = \mathbf{1}^\top x/\sqrt{d}$ alone, so the classes reduce to the scalar sets $f_{\mathrm{dc}}(X_\pm)$, whose hulls are the intervals $I_\pm$. By Problem 4.1 the max-margin is half the gap between the nearer endpoints, $\frac{1}{2}\,\mathrm{dist}(I_+, I_-)$. Identifying the projected line with $\mathrm{span}(\mathbf{1})$ gives the convex-hull form. □

Remark 6.2 (What this recovers). For the single white dot $\pm e_j$ of the base paper, $f_{\mathrm{dc}}(\pm e_j) = \pm 1/\sqrt{d}$, so the class gap is $2/\sqrt{d}$, equivalently the max-margin is $\gamma_{\mathrm{inv}} = 1/\sqrt{d}$ under the half-gap convention of Theorem 6.1. The unconstrained margin on that two-image set is 1, so the ratio is $1/\sqrt{d}$: invariance can cost a factor $\sqrt{d}$ in margin when the signal is purely localized.

Research move: Quotient out the nuisance

Imposing invariance is the same as projecting the data onto the invariant subspace and solving the ordinary problem there. The constraint did not make the problem harder; it made the data lower-dimensional. This "invariant problem = ordinary problem on the projected data" move recurs for general finite groups (replace Π_{inv} by P_G) and is the reason the convex-hull picture keeps working.

Problem 6.2 (General finite group)**.** Generalize: for a finite orthogonal group $\{R_g\}$ with average projector P_G, show the invariant linear max-margin of $X_\pm$ equals $\frac{1}{2}\,\mathrm{dist}(\mathrm{conv}\, P_G X_+, \mathrm{conv}\, P_G X_-)$.

Hints (peek one at a time):

- Repeat Problem 6.1(a) with R_g in place of S_s: invariance $\Leftrightarrow w \in V^G$.

- For $w \in V^G$, $w^\top x = w^\top P_G x$ (since $w = P_G w$ and P_G is self-adjoint), so the classifier only sees $P_G x$.

- Apply Theorem 4.1 to the projected sets.

Solution. Invariance of $w^\top x$ for all x forces $R_g^\top w = w$ for all g, i.e. $w \in V^G$. For such w, $w^\top x = (P_G w)^\top x = w^\top P_G x$, so every invariant linear classifier on $X_\pm$ is an ordinary linear classifier on $P_G X_\pm$ with the same normal and signed margins. Theorem 4.1 on $A = P_G X_+, B = P_G X_-$ gives the result. The cyclic case is $P_G = \Pi_{\mathrm{inv}}$. □

7 The η/L certificate, its failure of converse, and the two-sided bracket

The governing scalar of the whole project is η/L: the margin η of an invariant feature divided by its Lipschitz constant L. By Definition 2.2 it certifies $r_2 \geq \eta/L$. The subtle and important point is how tight that is.

Problem 7.1 (No universal converse)**.** Exhibit an invariant feature and a data point where the true robust radius is arbitrarily larger than η/L, so no inequality $r_2 \le C\,\eta/L$ can hold with a universal constant. (Hint: use a feature that depends only on the DC coordinate but is very flat near the boundary.)

Hints (peek one at a time):

- Work on the DC line $t = f_{\rm dc}(x)$; take the invariant feature $\Psi_n(x) = t^{2n+1}$.
- Put $x_\pm = \pm e$ with $e = d^{-1/2}\mathbf{1}$, labels ± 1. Compute the feature margin η and the Lipschitz constant of $t \mapsto t^{2n+1}$ on $[-1, 1]$.
- The decision boundary is $t = 0$, at Euclidean distance 1 from e. Compare to η/L.

Solution. With $g_n = \Psi_n$ and $x_\pm = \pm e$, the feature margin is $\eta = 1$. On $[-1, 1]$, $|\Psi_n'(t)| = (2n+1)t^{2n}$ peaks at $L_n = 2n+1$, so $\eta/L_n = 1/(2n+1) \to 0$. But the boundary $t = 0$ is at Euclidean distance $\|e - 0\|_2 = 1$ from e, and moving within $V_{\rm ac}$ does not change t at all, so $r_2(g_n, e) = 1$. The ratio $r_2/(\eta/L_n) = 2n+1$ is unbounded. Hence η/L is a certificate and a predictor, not a characterization, in general. □

Find-the-question: when a converse could hold The counterexample worked because the feature went *flat* near the boundary (its slope, the lower Lipschitz rate, fell to zero). Before reading on, ask: *what extra quantity would prevent this?* If the score were guaranteed to change at least at rate α along the path to the boundary, what radius upper bound would follow? You are about to invent the co-Lipschitz constant.

Theorem 7.1 (Two-sided bracket: a conditional converse)**.** *Suppose g is correct at x with $\mu = y\,g(x) > 0$, is L-Lipschitz toward the boundary, and admits a path γ to the boundary along which $|g(\gamma(t)) - g(x)| \ge \alpha\|\gamma(t) - x\|$ (a* co-Lipschitz*, or* lower-Lipschitz*, bound). Then $\mu/L \le r_2 \le \mu/\alpha$ (the score margin μ here plays the role of the feature margin η, so this is the η/L certificate completed to a bracket), with condition number $\kappa = L/\alpha \ge 1$; the certificate is exact ($r_2 = \mu/L$, $\kappa = 1$) for linear invariant features.*

Problem 7.2 (Prove the bracket)**.** Prove both arms, show $\kappa \ge 1$, and check the linear case gives $\kappa = 1$. Then reconcile with Problem 7.1: which constant blows up there?

Hints (peek one at a time):

- Lower arm is Definition 2.2.
- Upper arm: at the first boundary hit $\gamma(T)$, $g = 0$, so $\mu = |g(x) - g(\gamma(T))| \ge \alpha\|\gamma(T) - x\|$.
- The boundary point is a witness, so $r_2 \le \|\gamma(T) - x\| \le \mu/\alpha$.
- Linear $g = w^\top x$: along $\pm w$ the secant slope equals $\|w\|_2$, so $L = \alpha = \|w\|_2$.
- In Problem 7.1, the secant co-Lipschitz is $\alpha = 1$ but $L = 2n+1$, so $\kappa = 2n+1$ is what is unbounded.

Solution. Lower arm: no perturbation under μ/L flips the sign, so $r_2 \ge \mu/L$. Upper arm: at the boundary hit $g(\gamma(T)) = 0$, so $\mu = |g(x)| = |g(x) - g(\gamma(T))| \ge \alpha\|\gamma(T) - x\|$, giving $\|\gamma(T) - x\| \le \mu/\alpha$; since $\gamma(T)$ flips the prediction, $r_2 \le \mu/\alpha$. The secant slope cannot exceed the Lipschitz bound, so

$\alpha \le L$ and $\kappa \ge 1$. For linear g the straight path to the hyperplane has $L = \alpha = \|w\|_2$, so $\kappa = 1$ and $r_2 = \mu/\|w\|_2$ exactly. In the cubic example the straight DC path has $\alpha = 1$ (the feature drops from 1 to 0 over distance 1) while $L = 2n+1$; the bracket $[1/(2n+1), 1]$ correctly contains $r_2 = 1$, at the μ/α end, and the failure of a universal converse is exactly the unboundedness of κ over the model class. ☐

Research move: Bracket, do not chase equality

When an inequality has no converse, the move is not to give up but to name the missing ingredient and bound the gap by it. Here the gap is a condition number $\kappa = L/\alpha$. This both rescues η/L as an approximate characterization and explains the counterexample as the $\kappa \to \infty$ corner. The same instinct, "what one constant controls the gap," recurs in the lazy-regime theorem, where the gap is the orbit-variance.

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8 A convolution-square-pool layer computes the power spectrum

Linear invariance is weak (Theorem 6.1). The escape is a nonlinear invariant feature an actual network can compute.

Lemma 8.1 (Power-spectrum span)**.** *A single layer of circular convolution, pointwise squaring, and global sum-pooling realizes exactly the shift-invariant quadratic features, and these span the power-spectrum coordinates $|\hat{x}_k|^2$. On the ball $\|x\|_2 \le B$ the map $Q(x) = (|\hat{x}_k|^2)_k$ is $2B$-Lipschitz.*

Problem 8.1 (Prove the span and the Lipschitz bound)**.** (a) Show $\sum_j (w \star x)_j^2$, with $w \star x$ the circular cross-correlation, is shift-invariant and equals a weighted sum of $|\hat{x}_k|^2$. (b) Show $\|Q(x) - Q(y)\|_2 \le 2B\|x - y\|_2$ on the ball.

Hints (peek one at a time):

- (a) Circular correlation is diagonal in Fourier: $\widehat{w \star x}_k = \overline{\hat{w}_k}\,\hat{x}_k$. Use Parseval.
- (a) $\sum_j (w \star x)_j^2 = \sum_k |\hat{w}_k|^2 |\hat{x}_k|^2$, a nonnegative combination of power coordinates; varying w sweeps the coordinates.
- (b) For complex z, z', $\big||z|^2 - |z'|^2\big| \le |z - z'|(|z| + |z'|)$; apply coordinatewise to $\hat{x}, \hat{y}$ and use unitarity.

Solution. (a) By the convolution theorem and Parseval, $\sum_j (w \star x)_j^2 = \sum_k |\widehat{w \star x}_k|^2 = \sum_k |\hat{w}_k|^2 |\hat{x}_k|^2$; this is shift-invariant (each $|\hat{x}_k|^2$ is) and is a nonnegative-weighted sum of power coordinates, so a bank of such units spans the power spectrum. (b) Coordinatewise $\big||\hat{x}_k|^2 - |\hat{y}_k|^2\big| \le |\hat{x}_k - \hat{y}_k|(|\hat{x}_k| + |\hat{y}_k|)$; summing, Cauchy–Schwarz and unitarity of the DFT give $\|Q(x) - Q(y)\|_2 \le \|x - y\|_2(\|x\|_2 + \|y\|_2) \le 2B\|x - y\|_2$. ☐

Remark 8.2 (Phase-blindness and the bispectrum escape). The power spectrum is invariant but cannot separate signals that differ only in phase, such as a spike from its reflection (both have flat $|\hat{x}_k|^2$). The degree-three bispectrum $B_{k_1,k_2} = \hat{x}_{k_1}\hat{x}_{k_2}\overline{\hat{x}_{k_1+k_2}}$ is shift-invariant and phase-retaining, and separates the $\pm e_j$ dot with margin of order $d^{-3/2}$ in a single bispectrum coordinate (the full bispectrum vector in Euclidean norm scales differently), at the cost of a larger Lipschitz constant, hence a smaller certified η/L. This is the "go to higher degree" answer to the find-the-question prompt in §3.

9 Orbit-bundle form, orbit rank, and the degeneracy correction

Two structural facts make the optimization analysis possible and also explain why the textbook examples mislead.

Lemma 9.1 (Orbit-bundle form)**.** *A convolution-plus-global-pooling channel equals a two-layer network whose hidden units are tied into a shift orbit with an orbit-averaged readout:* $f(x) = \sum_r a_r \frac{1}{d} \sum_s \sigma(\langle w_r, S_s x\rangle)$*. Imposing exact shift-invariance on a two-layer ReLU network is exactly this weight-tying.*

Lemma 9.2 (Rank of an orbit)**.** *The cyclic orbit* $\{S_s u\}_s$ *spans a subspace of dimension equal to the number of nonzero Fourier coefficients of the base* u*.*

Problem 9.1 (Prove the rank formula and find the two degenerate bases)**.** (a) Prove Lemma 9.2. (b) Show that the two "natural" single-orbit constructions are each degenerate in opposite ways: a single cosine of frequency k, and a localized spike e_j.

Hints (peek one at a time):

- (a) In Fourier coordinates a shift multiplies coordinate k by a phase; the orbit lives in the span of the frequencies where $\hat{u}_k \neq 0$, and the distinct phase vectors are linearly independent there.
- (b) A pure cosine has only two nonzero Fourier coefficients (at $\pm k$): orbit rank 2, and its signed mean is zero.
- (b) A spike has all coefficients nonzero (full rank, orthonormal orbit) but flat power spectrum, so the power-spectrum invariant cannot separate two spike orbits.

Solution. (a) Write $u = \sum_{k:\hat{u}_k \neq 0} \hat{u}_k v_k$ with v_k the Fourier modes; $S_s u = \sum_k \hat{u}_k e^{-2\pi i s k/d} v_k$. The vectors $\{S_s u\}_s$ therefore lie in $\operatorname{span}\{v_k : \hat{u}_k \neq 0\}$, and the Vandermonde structure of the phases makes them span it, so the rank is $\#\{k : \hat{u}_k \neq 0\}$. (b) A cosine $\cos(2\pi k j/d)$ has nonzero coefficients only at $\pm k$: rank 2, far from the near-orthogonal $\Omega(d)$-cluster geometry the feature-averaging theorems assume, and its signed orbit mean (the direction those theorems use) is zero. A spike has all nonzero coefficients, so its orbit is full rank and orthonormal (the cluster assumption holds), but its power spectrum is flat, so $\mathrm{Sep}(Q) = 0$ and the power-spectrum (quadratic) invariant classifier cannot separate two spike classes (a higher-degree invariant such as the bispectrum still can; see §8). Neither base is a valid test of "does invariance re-point the implicit bias"; generic full-rank, phase-rich bases are. ☐

Research move: Find the degenerate case before you trust the example

The textbook constructions are exactly the measure-zero cases where the premise quietly fails. Before importing a theorem (here, feature-averaging non-robustness), check that your data satisfies its hypotheses; the rank formula is the cheap diagnostic that the single-orbit examples do not.

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10 Self-symmetrization: invariance is margin-free on orbit-closed data

Now the first optimization-side theorem. It is a clean, surprising negative result: on orbit-closed data, hard-wiring shift-invariance does not change the achievable margin at all.

Theorem 10.1 (Self-symmetrization)**.** *Let the data be orbit-closed with labels constant on orbits. For two-layer ReLU networks $f(x) = \sum_r a_r \sigma(w_r^\top x)$ (bias-free, degree-2 homogeneous), the minimum parameter norm $\frac{1}{2}\sum_r (a_r^2 + \|w_r\|^2)$ achieving unit margin is the same for the unconstrained class and the shift-invariant (orbit-tied) class. Imposing exact shift-invariance changes neither the max-margin value nor the achievable margin.*

Problem 10.1 (Prove self-symmetrization)**.** Two inequalities. (a) The easy direction. (b) The reverse: take any unconstrained solution, symmetrize it, and show the parameter norm does not grow. Use Problem 5.1.

Hints (peek one at a time):

- (a) The tied class is a subset of the unconstrained class, so its min-norm is at least as large.
- (b) Given an unconstrained $f^\star$ with margin ≥ 1, define $f_{\mathrm{sym}}(x) = \frac{1}{d}\sum_s f^\star(S_s x)$. Why is it invariant, in the tied class, and still margin ≥ 1?
- (b) Margin: orbit-closure means each $S_s x_i$ is a training point of the same label, so the average stays ≥ 1.
- (b) Norm: f_{sym} is a bundle of units $\{(a_r/d,\ S_{-s}w_r)\}$; sum the per-unit min-norm $|a_r/d|\,\|w_r\|$ over the d shifts using Problem 5.1.

Solution. (a) Tied $\subseteq$ unconstrained, so min-norm$_{\mathrm{tied}} \geq$ min-norm$_{\mathrm{unc}}$. (b) Let $f^\star$ be unconstrained with margin ≥ 1. Its orbit average $f_{\mathrm{sym}}(x) = \frac{1}{d}\sum_s f^\star(S_s x)$ is shift-invariant (reindex the pooling sum) and lies in the tied class by Lemma 9.1. Orbit-closure with constant labels gives $y_i f^\star(S_s x_i) \geq 1$ for all s, so $y_i f_{\mathrm{sym}}(x_i) \geq 1$: margin preserved. By Problem 5.1 each unit's min-norm is $|a|\,\|w\|$. The bundle realizing f_{sym} has units $\{(a_r/d, S_{-s}w_r)\}_{r,s}$ whose total min-norm is $\sum_r \sum_s \frac{|a_r|}{d}\|w_r\| = \sum_r |a_r|\|w_r\|$ (each of the d shifts contributes $\frac{|a_r|}{d}\|w_r\|$, and shifts preserve $\|w_r\|$), equal to the min-norm of $f^\star$. Hence min-norm$_{\mathrm{tied}} \leq$ min-norm$_{\mathrm{unc}}$, and the two are equal. □

Remark 10.2 (What it does and does not give). Self-symmetrization explains the experimental finding that on generic orbit data the unconstrained and tied networks reach the same robustness: gradient descent "self-symmetrizes." It does *not* fix which max-margin solution is selected, and the certified radius η/L depends on the Lipschitz term L, not only the margin. That selection is where invariance can still help, on cluster geometry; see §12.

11 The two-orbit quadratic surrogate has an exact margin

Theorem 11.1 (Two-orbit quadratic surrogate)**.** *For two-orbit data $X_+ = \mathcal{O}(u)$, $X_- = \mathcal{O}(v)$ with distinct power spectra $Q(u) \neq Q(v)$, the shift-invariant quadratic surrogate has max-margin $\eta_Q = \frac{1}{2}\|Q(u) - Q(v)\|_2$ and, on $\|x\|_2 \leq B$, certified radius $\geq \eta_Q/(2B)$.*

Problem 11.1 (Derive the exact margin)**.** Show that in power-spectrum feature space each class collapses to a single point, and apply convex-hull duality and Lemma 8.1.

Hints (peek one at a time):

- The power spectrum is shift-invariant, so every member of $\mathcal{O}(u)$ maps to the same point $Q(u)$.
- Two single points: the max-margin is half their distance (Theorem 4.1).

- The radius certificate is Lemma 8.1's $2B$-Lipschitz bound through Definition 2.2.

Solution. By shift-invariance $Q(S_s u) = Q(u)$, so X_+ and X_- map to the singletons $\{Q(u)\}$ and $\{Q(v)\}$. The maximum margin between two distinct points is half their Euclidean distance, $\eta_Q = \frac{1}{2}\|Q(u) - Q(v)\|_2$, attained by the bisecting hyperplane. Lemma 8.1 gives the feature map Lipschitz constant $2B$ on the ball, so Definition 2.2 yields radius $\geq \eta_Q/(2B)$. (One checks the optimal head lies in the real-filter span, so the surrogate is realizable.) □

12 The lazy-regime cluster theorem

This is the positive optimization result: in the lazy regime, on cluster-geometry data, the invariant network reaches an η/L at least as large as the unconstrained one, and strictly larger when the unconstrained interpolant carries non-invariant energy. The proof is the non-expansiveness of Problem 3.1 transported to the tangent kernel's function space.

Theorem 12.1 (Invariance lowers the Lipschitz constant in the lazy regime)**.** *In the lazy/NTK regime, with a smooth activation so the tangent feature map Φ is L_Φ-Lipschitz, on orbit-closed orbit-labeled data with a G-invariant tangent kernel: the orbit-averaging operator Π_G is the orthogonal projection of the tangent RKHS $\mathcal{H}_K$ onto its invariant subspace, hence non-expansive; the invariant min-norm interpolant satisfies $\|f_{\mathrm{inv}}\|_K \leq \|f_{\mathrm{unc}}\|_K$ at equal margin; therefore its RKHS-norm-based Lipschitz bound, hence the certified η/L, is no worse, and strictly better when the RKHS-norm-squared gap $\|(I - \Pi_G)f_{\mathrm{unc}}\|_K^2$ (the orbit-variance) is positive. This gap is the RKHS-norm gap; it bounds, but need not equal, the exact robust-radius or η/L gap.*

Problem 12.1 (Assemble the lazy-regime proof)**.** You are given: (i) lazy $\Rightarrow$ the trained net is the min-RKHS-norm interpolant of a fixed kernel; (ii) the RKHS Lipschitz bound $|f(x) - f(y)| \leq \|f\|_K \|\Phi(x) - \Phi(y)\|$. Build the theorem in three steps, then state the one place the smooth-activation hypothesis is load-bearing.

Hints (peek one at a time):

- Step 1: Π_G is self-adjoint and idempotent on $\mathcal{H}_K$ (it is the Reynolds projector, Problem 3.1), so it is an orthogonal projection: $\|f\|_K^2 = \|\Pi_G f\|_K^2 + \|(I - \Pi_G)f\|_K^2$.
- Step 2: orbit-constant labels make $\Pi_G f_{\mathrm{unc}}$ still interpolate with the same margin, so it is feasible for the invariant min-norm problem; hence $\|f_{\mathrm{inv}}\|_K \leq \|\Pi_G f_{\mathrm{unc}}\|_K \leq \|f_{\mathrm{unc}}\|_K$.
- Step 3: smaller RKHS norm gives smaller Lipschitz constant via (ii) and the L_Φ-Lipschitz feature map; at equal margin η/L improves.
- Load-bearing: step 3 needs Φ Lipschitz. The bare two-layer ReLU NTK feature map is *not* Lipschitz (its RKHS has unit-norm functions of unbounded Lipschitz constant), so the clean statement needs a smooth activation.

Solution. Step 1: a G-invariant kernel makes Π_G self-adjoint and idempotent on $\mathcal{H}_K$ with spectrum $\{0, 1\}$, hence the orthogonal projection onto the invariant subspace, with the Pythagorean identity. Step 2: orbit-constancy gives $y_i f_{\mathrm{unc}}(g \cdot x_i) \geq 1$ for all g, so $y_i(\Pi_G f_{\mathrm{unc}})(x_i) \geq 1$; thus $\Pi_G f_{\mathrm{unc}}$ is feasible for the invariant min-norm problem, giving $\|f_{\mathrm{inv}}\|_K \leq \|\Pi_G f_{\mathrm{unc}}\|_K$, which Pythagoras bounds by $\|f_{\mathrm{unc}}\|_K$. Step 3: the RKHS Lipschitz bound with an L_Φ-Lipschitz Φ gives $\mathrm{Lip}(f) \leq L_\Phi \|f\|_K$;

substituting the norm inequality, $L_{\text{inv}} \le L_{\text{unc}}$ at equal margin, so η/L does not decrease, and the gap is the discarded non-invariant energy $\|(I-\Pi_G)f_{\text{unc}}\|_K^2$, the orbit-variance. The smoothness is needed only in step 3; for bare ReLU the feature map is non-Lipschitz, so one restricts to smooth activations or carries $\sup\|\nabla\Phi\|$ explicitly. The rich-regime ReLU trajectory remains open. □

Research move: Transport a finite-dimensional lemma to function space

The whole theorem is Problem 3.1(iii) "a projection cannot lengthen a vector" moved from $\mathbb{R}^d$ to the RKHS, plus the dictionary norm → Lipschitz. When a clean linear-algebra fact stalls because the objects are now functions, look for the Hilbert space where the same words are true. The honest scoping (smooth activation, lazy regime) is part of the result, not an apology.

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13 The open problem: which solution does rich-regime training select?

Self-symmetrization (§10) says invariance is margin-free on generic orbit data; the lazy-regime theorem (§12) says it lowers L on cluster geometry. The gap between them is the live research question.

Conjecture 13.1 (Trajectory-level selection)**.** *On nondegenerate cluster-geometry orbit data there is a regime (initialization scale, width) where rich-regime gradient flow on the shift-invariant network reaches a KKT point of strictly larger η/L than the unconstrained network, even though both attain perfect clean margin.*

The checklist to attack it (the project's "what to prove"):

(P1) Write the invariant max-margin (KKT) program on two-orbit data and characterize the selected η/L.

(P2) Show the unconstrained gradient flow reaches a strictly smaller η/L on the same data, *or* exhibit the minimal orbit model where the feature-averaging premise provably holds.

(P3) Combine into Conjecture 13.1: weight-sharing re-points the bias toward larger η/L.

(P4) Decide whether the non-robust-selection premise holds on *any* full-rank orbit model at all.

Find-the-question: the right minimal model The experiments show invariance helps on "near-orthogonal cluster geometry" but not on generic single orbits. Before reading the project notes, ask: *what is the smallest data model that is simultaneously (a) full-rank/phase-rich enough for the invariant feature to separate the classes and (b) cluster-like enough for the feature-averaging non-robustness to bite?* The two requirements pull in opposite directions; finding the model where both hold is most of the open problem.

Part III: How to find the questions yourself

Parts I and II trained you to follow and reconstruct. This part trains the harder skill: generating the question. We do it by reverse-engineering how each Part II result was actually found, then giving you open prompts with no stated question.

14 The anatomy of the project's results

Every result in Part II came from one of a small number of moves. Naming them lets you reuse them.

1. **Quotient out the nuisance** (§6). The question "how much margin survives invariance?" became "solve the ordinary problem on the projected data." Whenever a constraint is a symmetry, project and re-ask.

2. **Diagonalize the symmetry** (§3, §8). Hard statements about shifts became one-liners in Fourier coordinates. When one group acts, change to its eigenbasis before doing anything else.

3. **Bracket what has no converse** (§7). A one-sided certificate with no converse was rescued by naming the missing constant (α) and bounding the gap by $\kappa = L/\alpha$.

4. **Find the degenerate case** (§9). The textbook examples were exactly where the premise failed; the rank formula is the diagnostic that exposes it.

5. **Reduce to the per-unit atom** (§10). A theorem about whole networks was the one-unit homogeneity identity (Problem 5.1) summed over an orbit.

6. **Transport a finite lemma to function space** (§12). "A projection cannot lengthen a vector" became the entire lazy-regime theorem once moved to the RKHS.

15 Two modes of solving, and why the second is rare

There is a real difference between *following* a decomposition someone hands you and *producing* it. The first is pattern-completion; the second needs you to (i) notice that a question is even there, (ii) guess the right object to introduce, and (iii) test it on the smallest example before investing in a proof. The hint ladders in Part II quietly performed (ii) and (iii) for you. The exercises below withhold them.

Problem 15.1 (Generate the question, then answer it: a new group)**.** We did everything for cyclic shifts. Consider instead the group of *horizontal and vertical* circular shifts on a $d_1 \times d_2$ image, $\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2}$. No sub-question is stated. Decide what the right analogues of the DC line, the orbit rank, and the power spectrum are, state the version of Theorem 6.1 you expect, and prove the one you are most confident about.

Hints (peek one at a time):

- Which move applies first? (Diagonalize: the 2D DFT diagonalizes both shift directions at once.)
- The fixed subspace of all 2D shifts is still the constants; the orbit rank is the number of nonzero 2D Fourier coefficients; the power spectrum is $|\hat{x}_{k_1,k_2}|^2$.
- Theorem 6.1 goes through verbatim with Π_{inv} the 2D DC projector; the proof is unchanged because it only used that P_G projects onto the constants.

Problem 15.2 (Generate the question: a different threat)**.** The bracket theorem (§7) was stated for ℓ_2. Without being told what to prove, work out what changes under an ℓ_∞ threat, what the right co-Lipschitz constant is, and whether the linear case is still exact. State a conjecture and test it on $g(x) = w^\top x$.

Hints (peek one at a time):

- The certificate uses the dual norm (Problem 2.1); for ℓ_∞ that is the ℓ_1 gradient.
- The linear exact case becomes $r_\infty = \mu/\|w\|_1$; check this is the point-to-hyperplane distance in ℓ_∞.

Problem 15.3 (Generate the question: where could invariance *help* robustness more?)**.** Self-symmetrization says invariance is margin-free on generic orbit data, and the lazy theorem says it helps via L on cluster geometry. Pose the sharpest experiment you can that would distinguish "invariance helps through margin" from "through Lipschitz," design its arms, and predict the outcome from the theory before any run.

16 Taste: which questions are worth asking

Not every well-posed question is worth a month. The ones that paid off here shared three features: they had a *small testable instance* (a $d = 4$ orbit, a two-image set), they sat at a *contradiction in the literature* (invariance reported as both helping and hurting robustness), and their answer *changed a decision* (which architecture to trust, which certificate to report). When you generate questions in Part III, score each on those three before committing. A question with no small instance is usually premature; a question whose answer changes nothing is usually not worth the proof.

Part IV: Experiments and where to contribute

17 The experimental backbone (the part of the project you can run)

The theory is tested by a capacity-matched architectural dissection. Four architectures share a parameter count and differ only in the shift-invariance operator: a *standard* network, an *anti-aliased* network (a fixed low-pass filter before each subsampling), an *exactly invariant* network (adaptive polyphase sampling, invariant to all integer shifts), and a *shift-augmented* network (invariance learned from data, not built in). The measured quantities are the ones from Part II: shift-consistency, the ℓ_2 robust radius (a minimum-norm attack), and the η/L decomposition into margin and Lipschitz terms.

The headline empirical facts, which you should be able to predict from the theory:

- Under standard training, η/L predicts the robust radius while shift-consistency does not.
- Under adversarial training, shift-consistency *anti*-predicts robustness, and the threat-matched ratio $\eta/\|\nabla M\|_1$ predicts it; the exactly invariant arm is the least robust, through reduced margin.

A new collaborator's natural first experimental task is to reproduce one cell of this dissection and read its results file, then re-derive which Part II result each number tests.

18 Open problems ranked by accessibility

1. **(P1) Two-orbit invariant KKT in closed form** (§13). Self-contained, frequency-domain, a good first theory contribution. Start from Theorem 11.1 and replace the quadratic surrogate by ReLU.

2. **The minimal cluster-orbit model** (find-the-question, §13). Decide whether the feature-averaging premise can hold on any full-rank orbit model; if not, the conjecture must be restricted.

3. **Rich-regime selection** (Conjecture 13.1). The hardest: track which KKT point gradient flow selects when several exist. Needs trajectory information, not just the variational limit.

4. **ℓ_∞ bracket and tighter κ estimates** (Problem 15.2). Extend the two-sided certificate and estimate κ for the power-spectrum model.

19 Reading roadmap

1. This primer, Parts I–II, with the problems done closed-book.
2. The paper `main.tex` end to end; you should recognize every theorem from Part II.
3. The technical report `theory_full_checked_v2.tex` for the reference proofs and the optimization roadmap section.
4. One experimental cell reproduced; one open problem from Part IV attempted for two weeks before proposing.

Glossary

Robust radius $r_p(f, x)$
distance from x to the nearest point of opposite prediction, in ℓ_p.

Certificate
a guaranteed lower bound on the robust radius (versus an attack, an upper bound).

η/L
invariant-feature margin divided by its Lipschitz constant; certifies $r_2 \geq \eta/L$.

Co-Lipschitz constant α
a guaranteed lower rate of change toward the boundary; gives the converse arm.

Condition number $\kappa = L/\alpha$
controls how far η/L is from the true radius; $\kappa = 1$ is exact.

Orbit
the set of all shifts of a signal; orbit-closed means closed under all shifts.

DC / AC subspace
the constant line span($\mathbf{1}$) and its zero-sum complement.

Power spectrum / bispectrum
degree-two phase-blind, and degree-three phase-retaining, shift invariants.

Reynolds projector P_G
the group average $|G|^{-1} \sum_g R_g$; the orthogonal projector onto the invariants.

Self-symmetrization
on orbit-closed data, gradient descent reaches an invariant max-margin solution on its own.

Lazy / rich regime
very-wide fixed-kernel training versus small-init feature-learning training.

Feature averaging
the implicit-bias mechanism by which max-margin solutions become non-robust on cluster data.

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Colophon

This primer is a teaching companion to the paper and the technical report; all theorem statements and proofs are the verified versions in `theory_full_checked_v2.tex`. It follows the project's house writing rules. Corrections and better worked examples are welcome and should be checked against the technical report before they are added.